In [1]:

```python
import pandas as pd
import warnings

warnings.filterwarnings("ignore")

from sklearn.ensemble import BaggingClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier

```

In [5]:

```python
#example2
data = {
    "Gender":["Female" ,"Male", "Male", "Female", "Male"],
    "Age":[33,57,41,49,36],
    "Income":['480000','320000','900000','540000','450000'],
    "BMI":[22.1,29.5,24.0,31.2,27.8],
    "Illness":["No","Yes","No","Yes","No"],
}

df = pd.DataFrame(data)
print(df)

```

```
   Gender  Age  Income   BMI Illness
0  Female   33  480000  22.1      No
1    Male   57  320000  29.5     Yes
2    Male   41  900000  24.0      No
3  Female   49  540000  31.2     Yes
4    Male   36  450000  27.8      No
```

In [6]:

```python
X = df[["Age", "BMI"]]
y = df["Illness"]
```

In [7]:

```python
# Base model
base_tree = DecisionTreeClassifier()

# Bagging classifier
bagging_model = BaggingClassifier(
    estimator=base_tree,
    n_estimators=10,
    bootstrap=True,
    random_state=42
)

# Train model
bagging_model.fit(X, y)
```

Out[7]:

```
▾ BaggingClassifier
BaggingClassifier(estimator=DecisionTreeClassifier(), random_state=42)
▾ estimator: DecisionTreeClassifier
DecisionTreeClassifier()
▸ DecisionTreeClassifier
```

In [14]:

```
# New person data
new_person = [[33, 22.1]]

prediction = bagging_model.predict(new_person)
print("Bagging Prediction:", "Yes illness " if prediction[0] == 1 else "No illes
```

```
Bagging Prediction: No illess
```

In [15]:

```
rf_model = RandomForestClassifier(
    n_estimators=10,
    random_state=42
)

# Train Random Forest
rf_model.fit(X, y)
```

Out[15]:

```
RandomForestClassifier

RandomForestClassifier(n_estimators=10, random_state=42)
```

In [17]:

```
rf_prediction = rf_model.predict(new_person)
print("Bagging Prediction:", "Yes illness" if rf_prediction[0] == 1 else "No ill
```

```
Bagging Prediction: No illess
```

In [ ]:

In [ ]: